

SMART Optical Network Integrating PTMP Sensing, 10×1-Tb/s Metro Superchannel, and Fixed-Mobile Access

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Abstract We present SMART (*Sensing-Metro-Access-RoF-Transport*) optical network integrating, for the first time, point-to-multipoint multi-event forward sensing with sub-28-m resolution, 10×1-Tb/s metro superchannel, co-wavelength access of super-rated 200G coherent PON with 35-dB power budget (ITU-T E2-class) and 1.05-Tb/s CPRI-equivalent rate 16384-QAM RoF. © 2026 The Author(s)

Introduction

Optical networks are widely deployed infrastructures for massive data transmission, exchange, and distribution. In the AI era, emerging applications demand not only enhanced capacity, but also stringent requirements on latency, flexibility, and service agility for accessing both data streams and computing resources. Coherent optics offer significant advantages over conventional intensity-modulation direct-detection solution in terms of bitrate, reach, and flexibility, providing a physical-layer foundation for all-optical metro-access networks without time-consuming optical-electrical-optical conversions [1,2].

More importantly, coherent detection enables full extraction of the optical field, including phase information, naturally supporting integrated sensing and communications. This ability transforms optical networks from capacity-driven transport pipes into function-driven intelligent infrastructures, as depicted in Fig. 1(a).

However, enabling sensing in a point-to-multipoint (PTMP) access network from the central office remains a long-standing challenge [3]. As shown in Fig. 1(c), in conventional backscattering-based distributed acoustic sensing (DAS), colorless passive splitters indiscriminately combine backscattered signals from multiple distribution fibers, preventing unambiguous identification and localization of vibration events. This intrinsic ambiguity cannot be resolved by frequency- or time-division multiplexing, posing a fundamental barrier to PTMP sensing in metro-access networks.

In this work, we propose and experimentally demonstrate a SMART (*Sensing-Metro-Access-RoF-Transport*) optical network with tightly integrated communication and sensing functionalities: (1) a PTMP multi-event forward sensing scheme is proposed for the first time, where a dedicated branch is used to decouple vibration

events occurring on the feeder and distribution fibers, achieving sub-28-m localization resolution over an >80-km fiber link with multiple same-frequency vibrations; (2) 10×1.0-Tb/s PS-64-QAM wavelength-division-multiplexed (WDM) metro superchannel is demonstrated over 160-km SSMF; (3) single-wavelength fixed-mobile converged access is realized via subcarrier multiplexing, simultaneously carrying a super-rated 240-Gb/s coherent PON with a 35-dB power budget and a 1.05-Tb/s common public radio interface (CPRI)-equivalent 16384-QAM RoF fronthaul enabled by a cascaded digital-analog RoF (CDA-RoF). The SMART network maximizes both capacity and functionality of the fiber infrastructure.

PTMP Sensing-based SMART Optical Network

We exploit residual optical carrier induced by modulator bias deviation to enable ISAC [Fig. 1(b)], serving both phase-related DSP [4] for coherent communications and forward vibration sensing and localization. Additionally, we use delayed self-heterodyne interferometer (DSHI) unit for PTMP sensing, as shown in Fig. 1(d). Without loss of generality, we assume two vibration events are located at the feeder and distribution fiber, respectively. In the far-end DSHI units, the acousto-optic modulators (AOM #1~4) shift the frequency of input optical carrier by $\Delta f_1, \Delta f_1 + \Delta f_2, \Delta f_3, \Delta f_3 + \Delta f_4$, for backscatter-tolerant bi-directional transmission [5,6]. At the OLT, phase $\phi_i(t)$ at IFs from DSHI 1&2 are extracted as

$$\phi_1(t) = \phi_{PN}(t - 2T) + \phi_{vib1}(t - 2T + \tau_A) + \phi_{vib1}(t - \tau_A) - \phi_{PN}(t), \quad (1)$$

$$\phi_2(t) = \phi_{PN}(t - 2T - \tau) + \phi_{vib1}(t - 2T - \tau + \tau_A) + \phi_{vib1}(t - \tau_A) - \phi_{PN}(t), \quad (2)$$

$$\phi_3(t) = \phi_{PN}(t - 2T - 2T_{ONU}) + \phi_{vib1}(t - 2T - 2T_{ONU} + \tau_A) + \phi_{vib1}(t - \tau_A) + \phi_{vib2}(t - 2T - 2T_{ONU} + \tau_B) + \phi_{vib2}(t - \tau_B) - \phi_{PN}(t), \quad (3)$$

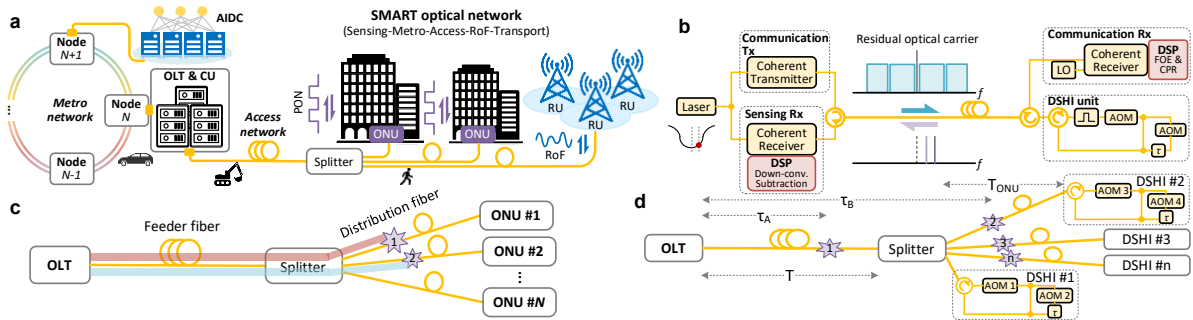


Fig. 1: (a) The vision of the SMART optical network. (b) Basic principle of a shared residual optical carrier for ISAC. (c) The challenge of multiple vibrations identification on different distribution fibers in a PTMP access network. (d) Illustration of forward sensing-based PTMP sensing scheme. DSHI: delayed self-heterodyne interferometer.

$\varphi_4(t) = \varphi_{PN}(t - 2T - 2T_{ONU} - \tau) - \varphi_{PN}(t)$
 $+ \varphi_{vib1}(t - 2T - 2T_{ONU} - \tau + \tau_A) + \varphi_{vib1}(t - \tau_A)$
 $+ \varphi_{vib2}(t - 2T - 2T_{ONU} - \tau + \tau_B) + \varphi_{vib2}(t - \tau_B),$ (4)

where φ_{PN} is the laser phase noise, $\varphi_{vib1/2}$ are vibration phase of event 1/2. T , τ_A , τ_B , and T_{ONU} are the propagation time from OLT to splitter, event 1, event 2, and ONU1, respectively. τ is the delay in DSHI. Then, we extract the phases obtained from Eq. (1)-(2) as

$$\begin{aligned}
 I_1(t) &= \varphi_1(t) - \varphi_2(t) \\
 &\approx \varphi_{vib1}(t - 2T + \tau_A) - \varphi_{vib1}(t - 2T - 2\tau + \tau_A), \quad (5) \\
 I_2(t) &= \varphi_1(t - 2\tau) - \varphi_2(t) \\
 &\approx \varphi_{vib1}(t - 2\tau - \tau_A) - \varphi_{vib1}(t - \tau_A). \quad (6)
 \end{aligned}$$

Here the laser phase noise terms $\varphi_{PN}(t - 2T) - \varphi_{PN}(t - 2T - \tau)$ is neglected, due to a relatively short delay fiber inside the DSHI. It is found that $I_1(t) = -I_2(t - 2T + 2\tau_A)$, indicating that performing cross correlation between $I_1(t)$ and $I_2(t)$ leads to localization of vibration event on the feeder fiber. Similarly, we repeat the process on the phases from Eq. (3) and (4) as

$$\begin{aligned}
 I_3(t) &= \varphi_3(t) - \varphi_4(t) \approx \varphi_{vib1}(t - 2T - 2T_{ONU} + \tau_A) \\
 &- \varphi_{vib1}(t - 2T - 2T_{ONU} - \tau + \tau_A) + \varphi_{vib2}(t - 2T - 2T_{ONU} + \tau_B) \\
 &- \varphi_{vib2}(t - 2T - 2T_{ONU} - \tau + \tau_B), \quad (7) \\
 I_4(t) &= \varphi_3(t - 2\tau) - \varphi_4(t) \approx \varphi_{vib1}(t - 2\tau - \tau_A) \\
 &- \varphi_{vib1}(t - \tau_A) + \varphi_{vib2}(t - 2\tau - \tau_B) - \varphi_{vib2}(t - \tau_B). \quad (8)
 \end{aligned}$$

Then we subtract the phases to decouple the vibration event on the feeder and distribution fibers as

$$\begin{aligned}
 D_1(t) &= I_3(t) - I_1(t - 2T_{ONU}) = \varphi_{vib2}(t - 2T - 2T_{ONU} + \tau_B) \\
 &- \varphi_{vib2}(t - 2T - 2T_{ONU} - \tau + \tau_B), \quad (9) \\
 D_2(t) &= I_4(t) - I_2(t) = \varphi_{vib2}(t - \tau - \tau_B) - \varphi_{vib2}(t - \tau_B). \quad (10)
 \end{aligned}$$

We can find that $D_1(t) = -D_2(t - 2T - 2T_{ONU} + \tau_B)$, meaning that cross correlation between $D_1(t)$ and $D_2(t)$ generates a localization peak corresponding to $2T + 2T_{ONU} - \tau_B$. Therefore, the proposed scheme can distinguish multiple events even with the same vibration frequency. Note that the architecture can be extended to tens of ONUs for PTMP network, provided (1) the AOMs in each DSHI unit converts the residual carrier to a different IF, and (2) delay fibers in all the DSHI have consistent values of τ .

Experimental Setup and DSP

Fig. 2 depicts the experimental setup of the SMART network. The metro transmitter consists of 10 external cavity lasers (ECLs) spaced at 110 GHz, which are combined and fed into a 27-GHz IQ modulator biased slightly above the null point to generate a residual carrier. A 120-GSa/s arbitrary waveform generator (AWG, Keysight Technologies M8194) produces a dual-band 2×45-GBd PS-64-QAM signal. The source entropy is 5.56 bit/symbol, corresponding to a line rate of 1.0 Tb/s per wavelength channel. The access channel is generated at the OLT/ONU and seeded by a 1-kHz laser. The AWG produces a converged fixed-mobile access signal via digital

subcarrier multiplexing, including a 2×18-GBd DA-RoF signal at baseband to benefit from higher electrical SNR and a 2×15-GBd 16-QAM signal in the high-frequency region for super-rated 200G TFDM-PON [7]. Polarization-division multiplexing (PDM) is emulated by signal splitting, ~10-ns decorrelation, and recombination using a polarization-beam combiner. The metro superchannel and access signal are then combined and power-flattened using a wavelength-selective switch (WSS). The total launch optical power is set as 13 dBm.

For the metro case, the WDM signals are transmitted over two 80-km spans of SSMF with EDFA amplification. At the metro receiver, the target wavelength is selected and detected using a 70-GHz standard coherent receiver with a 100-kHz local oscillator (LO). A 256-GSa/s real-time oscilloscope (RTO) is employed to capture the electrical waveforms for offline processing.

For the access case, the signal is separated by another WSS, amplified to 8 dBm, and transmitted over a 10-km SSMF for RoF fronthaul and 20-km SSMF for PON, respectively. The ONU/RU detects the access signal without using optical amplifiers, and the RTO sampling rate is reduced to 128 GSa/s for lower cost.

For PTMP forward sensing, a DSHI is placed after node 1 to monitor the metro link, while another DSHI is located at the ONU/RU to sense the distribution fibers. The delay inside DSHI is implemented by a 80-m fiber. Using circulators, the residual optical carrier is filtered and sent back to the OLT/CU, which is self-heterodyned with the transmitting laser to suppress laser phase noise. Notably, metro networks are typically deployed in a dual-fiber ring topology, where circulators are not required.

Experimental Results and Discussions

Event Separation: Fig. 3 evaluates the PTMP forward sensing performance. Two 20-kHz vibration events with a 10-ms period are introduced using two PZTs with lengths of 15 and 25 m. Fig. 3(a) and 3(b) shows the extracted phase after a digital low-pass filter (LPF), and subsequently by a [15, 40]-kHz band-pass filter to suppress out-of-band noise. As plotted in Fig. 3(a) and 3(b), events 1 and 2 are overlapped in both the time and frequency domains. By applying Eq. (5)-(10), the vibration phases of events 1 and 2 are separated in Fig. 3(c) and Fig. 3(d). Their locations are then obtained from the corresponding cross-correlation waveforms in Fig. 3(e) and Fig. 3(f).

Multi-event localization: The localization accuracy is further characterized under different network scenarios: (i) When event 1 is on the feeder fiber and event 2 is on the distribution fiber [Fig. 3(g)], the standard deviations (STDs) of the localization error are 0.52 and 28.70 m, respectively. The degraded accuracy of event 2 mainly results from the higher insertion loss (8 dB) of the 25-m PZT and partly by the residual crosstalk from event 1. (ii) When both events are on two distribution fibers [Fig. 3(j)],

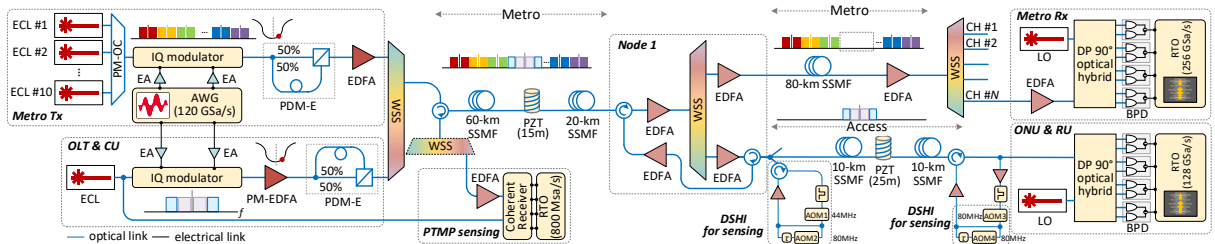


Fig. 2: Experimental setup. PM-OC: polarization-maintaining optical coupler; AWG: arbitrary waveform generator; EA: electrical amplifier; PM-EDFA: polarization-maintaining erbium-doped fiber amplifier; PDM-E: polarization-division-multiplexing emulator; WSS: wavelength-selective switch; PZT: piezoelectric transducer; DP: dual-polarization; BPD: balanced photodiode; RTO: real-time oscilloscope. AOM: acousto-optic modulator; DSHI: delayed self-heterodyne interferometer.

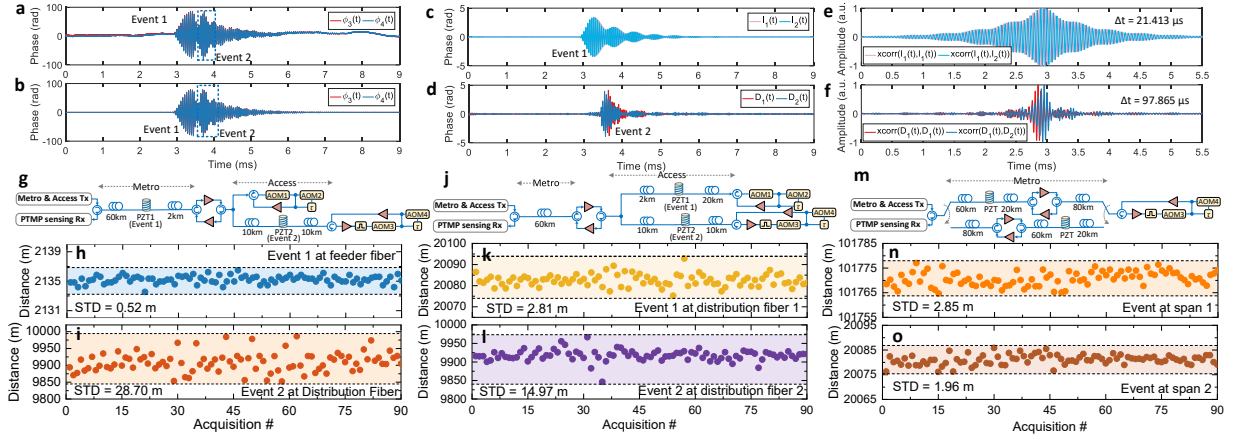


Fig. 3: (a) Measured phase after LPF. (b) Measured vibration phase after BPF. Recovered vibration phase of (c) event 1 and (d) event 2. Cross-correlation waveform of (e) event 1 and (f) event 2. (g) Setup and (h)-(i) vibration localization accuracy for 2 events on feeder and distribution fiber case. (j) Setup and (k)-(l) vibration localization accuracy for 2 events on different distribution fiber case. (m) Setup and (n)-(o) vibration localization accuracy for 1 event on different span of metro fiber case. STD: standard deviation.

the STDs are 2.81 and 14.97 m, respectively, where the accuracy degradation of event 1 is caused by the longer transmission distance. (iii) For a single event over a 160-km metro link using 15-m PZT [Fig.3(m)], the localization error STD remains below 3 m in Fig. 3(n) and 3(o), revealing high resolution.

Metro transmission: Fig. 4 shows the communication performance of metro and fixed-mobile access signals. Fig. 4(a) shows the optical spectra of the metro-access WDM superchannel, where an OSNR degradation of ~12 dB is observed after 160-km SSMF transmission. For metro signals, Fig. 4(b) plots the measured normalized generalized mutual information (NGMI) as a function of channel index, indicating that all WDM channels meet the 0.8798 NGMI threshold [8]. With a code rate of 0.8402, a net capacity of 8.3(=10×0.828) Tb/s is achieved. Fig. 4(c) presents the NGMI versus transmission distance for channel #5, where a ~0.03 NGMI penalty is mainly attributed to OSNR degradation.

PON access: Figs. 4(d) and 4(g) depict the filtered optical and electrical spectra after node 1. As shown in Fig. 4(e), modulation formats from QPSK to 32-QAM are evaluated to enable flexible-rate PON operation. Fig. 4(f)

illustrates the bitrate as a function of optical path loss (OPL), achieving a peak data rate of 300 Gb/s and a 35-dB power budget for 240-Gb/s 16-QAM transmission.

RoF Access: To concurrently support RoF fronthaul, DA-RoF modulation is loaded to the inner subcarriers. The recovered SNR and corresponding CPRI-equivalent rates are presented in Fig. 4(h), where a 2.11-Tb/s CPRI-equivalent rate is achieved with 256-QAM. By employing 3-order DA-RoF [9], the SNR is enhanced to 46.0 dB for 17.6-GHz-bandwidth 16384-QAM. Finally, Figs. 4(i)-4(l) show the recovered constellations for modulation formats from 256-QAM to 16384-QAM.

Conclusions

We demonstrate a SMART optical network that seamlessly integrates, for the first time, sub-28-m resolution PTMP sensing without modifying ODN in access network, a 10-Tb/s metro superchannel, and co-wavelength access supporting super-rated 200G PON at ITU-T E2-class [10] power budget together with >2-Tb/s CPRI-equivalent 256-QAM RoF. The architecture unifies physical-layer high-capacity transmission and high-resolution sensing, paving the way for future low-latency and intelligent optical networks.

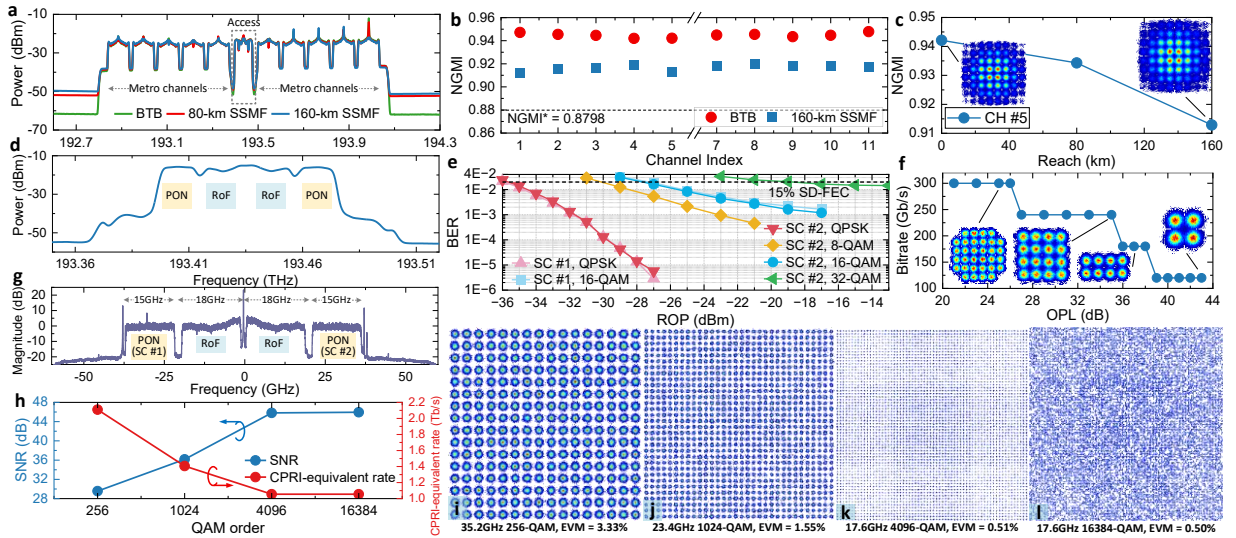


Fig. 4: (a) Measured optical spectra of metro-access superchannel. (b) Measured NGMI vs. channel index of metro signal at BTB and 160-km SSMF. (c) Measured NGMI vs. distance of metro signal. (d) Measured optical spectrum of access signal after node 1. (e) Measured BER vs. received optical power (ROP) of PON signal after 80+20-km SSMF. (f) Measured bitrate vs. optical path loss (OPL) of PON signal. (g) Received electrical spectrum of access signal. (h) Recovered SNR and CPRI-equivalent rate vs. QAM order of RoF signals after 80+10-km SSMF. (i)-(l) Recovered constellations of RoF signals from 256-QAM to 16384-QAM.

Acknowledgements

National Key Research and Development Program of China (2023YFB2906000), and Natural National Science Foundation of China (62475147, 62501017, 62271010, 62271305).

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